

RPT7050A

1.0 Specification References

Parameter	Description
a. Rakon part number	E8771LF
b. Description	19.44MHz RPT7050A TCXO
c. Package	L x W x H: 7.0 x 5.0 x 1.5 mm nom. (6 pad)



2.0 Absolute Maximum Ratings ¹

Parameter	Min.	Max.	Unit.
a. Junction temperature		150	°C
b. Supply voltage (Vcc)	-0.5	7	V
c. All other inputs	-0.5	Vcc + 0.5	V
d. Power dissipation		100	mW

3.0 Frequency Characteristics

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Nominal frequency		19.44		MHz	
b. Frequency calibration			±1	ppm	Initial accuracy at 25°C ±1°C
c. Reflow shift			±0.5	ppm	Pre to post reflow ΔF (measured ≥ 60 minutes after reflow)
d. Frequency stability over temperature			±2.5	ppm	Reference to (F _{MAX} + F _{MIN})/2
e. Operating temperature range	-40		+85	°C	
f. Supply voltage stability		±25		ppb	±5% variation, reference to frequency at nominal supply voltage
g. Load sensitivity		±50		ppb	±5pF, reference to frequency at nominal load
h. Long term stability (aging at 25°C)			±1 ±3	ppm ppm	1 year 10 years
i. Acceleration sensitivity		< 2		ppb/g	Gamma vector, 3-axes, 30-1500Hz
j. Start-up time			15	ms	90% amplitude

4.0 Power Supply

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Supply voltage (Vcc)	3.135	3.3	3.465	V	
b. Supply current		4	6	mA	

¹ Operating beyond this limit may result in change or permanent damage to the device.

5.0 Oscillator Output – HCMOS

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Output waveform					HCMOS (LVCMOS & LVTTTL compatible as per JESD8C)
b. Output voltage level low (V_{OL})			10% V_{CC}	V	
c. Output voltage level high (V_{OH})	90% V_{CC}			V	
d. Rise and fall time			8	ns	10% to 90% level
e. Duty cycle	45		55	%	At 50% level
f. Load		15		pF	

6.0 Tri-State Control ²

Parameter	Min.	Typ.	Max.	Unit	Test Condition / Description
a. Tri-state mode					The device features a tri-state mode which allows the output to be disabled and brought into a high impedance state
b. Tri-state control (pin 6), input level low (V_{IL})			20% V_{CC}	V	Device disabled (output in high impedance state)
c. Tri-state control (pin 6), input level high (V_{IH})	60% V_{CC}			V	Device enabled (operating)
d. Current when in tri-state mode		2		mA	
e. Output enable time			100	μs	

7.0 SSB Phase Noise (at 25°C)

Parameter	Typ.	Unit.	Test Condition / Description
a. 1Hz offset	-65	dBc/Hz	
b. 10Hz offset	-94	dBc/Hz	
c. 100Hz offset	-124	dBc/Hz	
d. 1kHz offset	-142	dBc/Hz	
e. 10kHz offset	-150	dBc/Hz	
f. 100kHz offset	-153	dBc/Hz	
g. 1MHz offset	-155	dBc/Hz	
h. RMS Phase Jitter	0.3	ps	Integrated 12kHz to 5MHz

8.0 Root Allan Variance (at 25°C)

Parameter	Typ.	Unit	Test Condition / Description
a. Root Allan Variance	1•10 ⁻¹⁰ 5•10 ⁻¹¹ 5•10 ⁻¹¹ 5•10 ⁻¹¹		tau = 0.2s tau = 1.0s tau = 10s tau = 100s

² The tri-state control (enable) input pin has an internal 100kΩ pull up resistor which allows it to be left unconnected if not used. When in tri-state mode, the output stage is disabled, but the oscillator and compensation circuit are still active.

9.0 Marking

Parameter	Description
a. Type	Laser marked
b. Line 1	[R X XX] R = Rakon, X XX = manufacturing identifier
c. Line 2*	[EXXXX YW] EXXXX = abbreviated part number, YW = device date code
d. Note	* Rotated by 180°

10.0 Manufacturing Information

Parameter	Description
a. Reflow soldering	Reflow profile as per IPC/JEDEC J-STD-020E (see drawing)
b. Packaging description	Tape & Reel as per EIA-481-E (see drawing)
c. Application note	For optimum performance, follow the instructions in Guidelines for use of Pluto®/Pluto+™ Ultra-Stable TCXOs

11.0 Environmental Specification

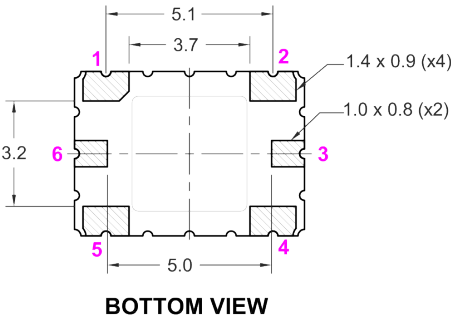
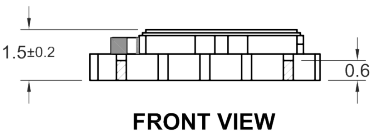
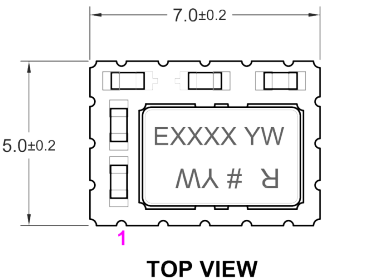
Parameter	Description
a. RoHS	Parts are fully compliant with the European Union directives 2011/65/EU and 2015/863/EU (amending Annex II to directive 2011/65/EU) on the restriction of the use of certain hazardous substances in electrical and electronic equipment
b. Solderability	JESD22-B102, M1, condition E (IPC/EIA J-STD-002A), 245°C for 5s, precondition for 16 hours at +150°C
c. High Temperature Operating Life (HTOL)	JESD22-A108, 1008 hours at +125°C
d. Temperature cycle	JESD22-A104, 500 cycles, -55°C to +125°C
e. Low temperature storage	IEC 60068-2-1 test Ab, 1000 hours at -55°C
f. High temperature storage	IEC 60068-2-2 test Bb, 1000 hours at +150°C
g. Moisture resistance	JESD22-A113, MSL = 1
h. Temperature / Humidity bias	JESD22-A101, 1008 hours at +85°C / 85% R.H., precondition: 3 Reflow cycles (peak temperature 260°C)
i. Mechanical vibration	JESD22-B103, 20g, 60-2000Hz, 4 hours in each of three axes (12 hours total)
j. Mechanical shock	JESD22-B104, 1500g _n , 0.5ms, 5 pulses in each of 6 directions
k. Ageing	MIL-PRF-55310, 1008 hours at +85°C, precondition: 3 Reflow cycles (peak temperature 260°C)
l. Resistance to soldering heat	IPC/JEDEC J-STD-020, 3 reflow cycles (peak temperature 260°C)

12.0 Disclaimer

Parameter	Description
a. Disclaimer	"Samples supplied according to this specification are supplied from our development or pre-production programme and as such are not qualification approved products. No condition, warranty or representation regarding the quality, suitability, performance, life or continuation of supply is given or implied and Guarantee in clause 6.1 of our Standard Conditions of Sale is not applicable. The right is reserved to change the design or specification or cease supply without notice." Rakon Limited

13.0 Model Outline:

MODEL DRAWING



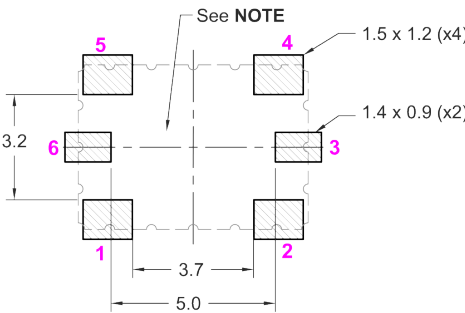
Pin Connections

Pin	Connections
1	Do not connect (can be connected to GND)
2	GND
3	Do not connect
4	RF Output
5	Supply Voltage (Vcc)
6	Tri-state Control (Enable) ²

NOTE:

- The area between the pads is a keep-out area, no tracks or ground plane allowed on any layer.

RECOMMENDED PAD LAYOUT - TOP VIEW



TITLE: RPT7050 MODEL (6 Pad)

RELATED DRAWINGS:

FILENAME: CAT760

REVISION: F

DATE: 13-Oct-2020

SCALE: 5 : 1

Millimetres

TOLERANCES:

XX =

X.X = ±0.2

X.XX = ±0.10

X.XXX =

X° =

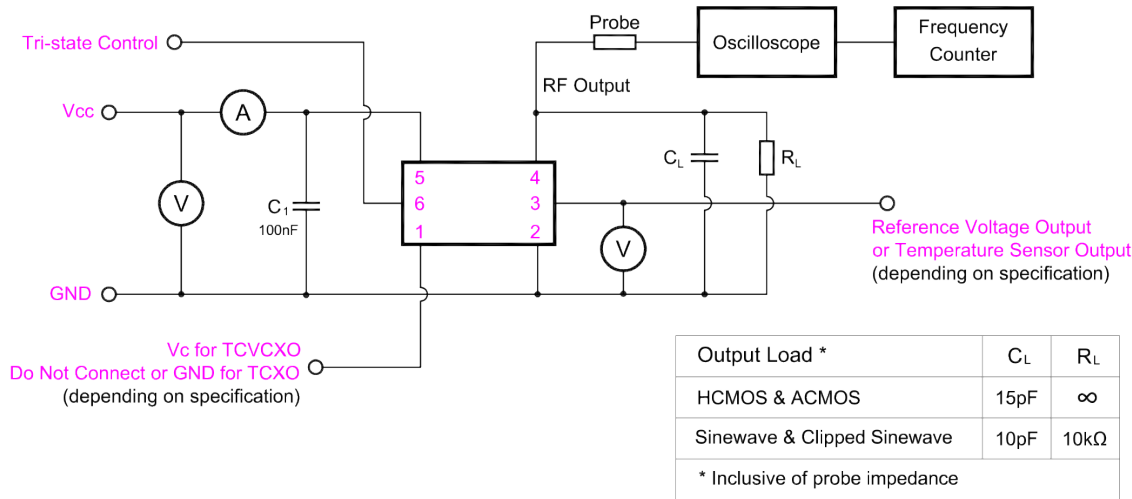
Hole =

rakon

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14.0 Test Circuit and Output Waveform:

6 PIN TEST CIRCUIT:



TITLE: Pluto+ TCXO/TCVCXO TEST CIRCUIT

FILENAME: CAT781

RELATED DRAWINGS:

REVISION: A

DATE: 20-Mar-13

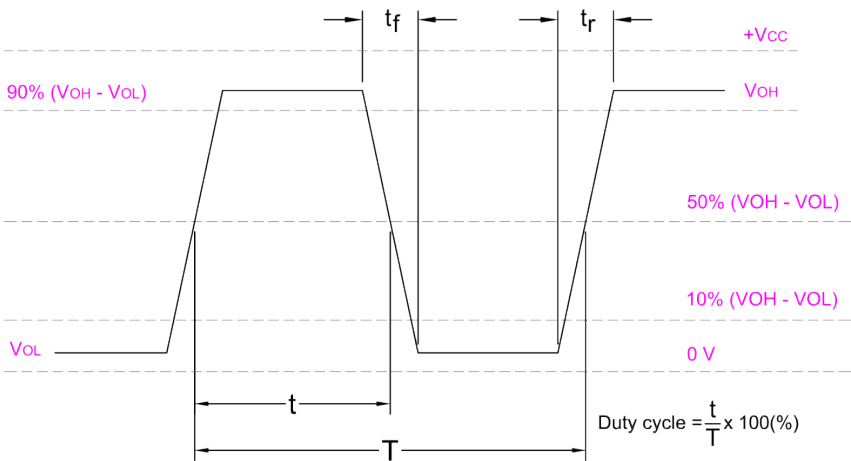
SCALE: NTS

Millimetres

rakon

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OUTPUT WAVEFORM - HCMOS & AC MOS:



TITLE: Pluto+ TCXO/TCVCXO OUTPUT WAVEFORM

FILENAME: CAT784

RELATED DRAWINGS:

REVISION: A

DATE: 15-Mar-13

SCALE: NTS

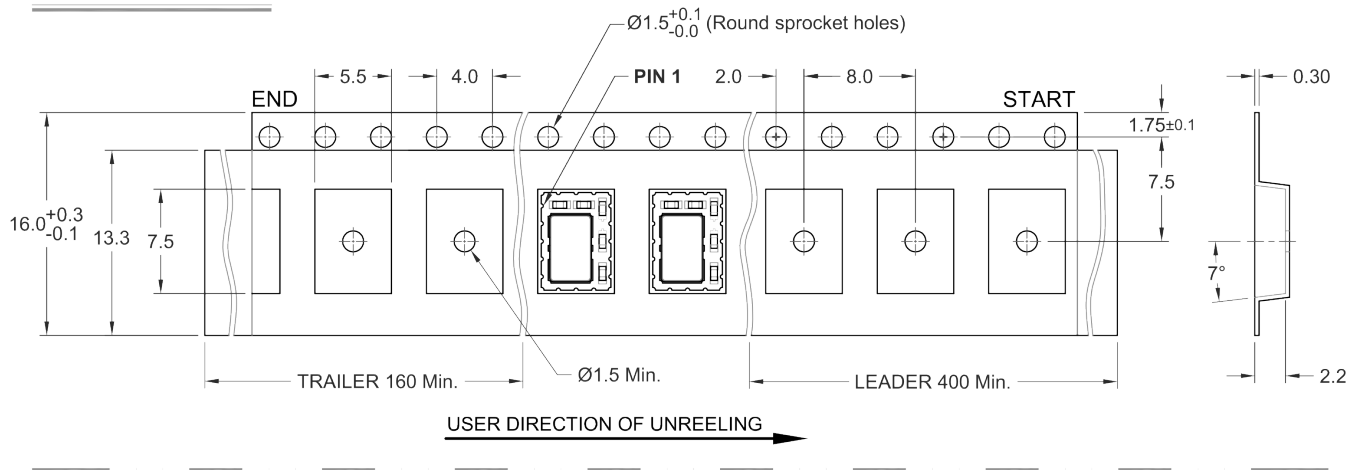
Millimetres

rakon

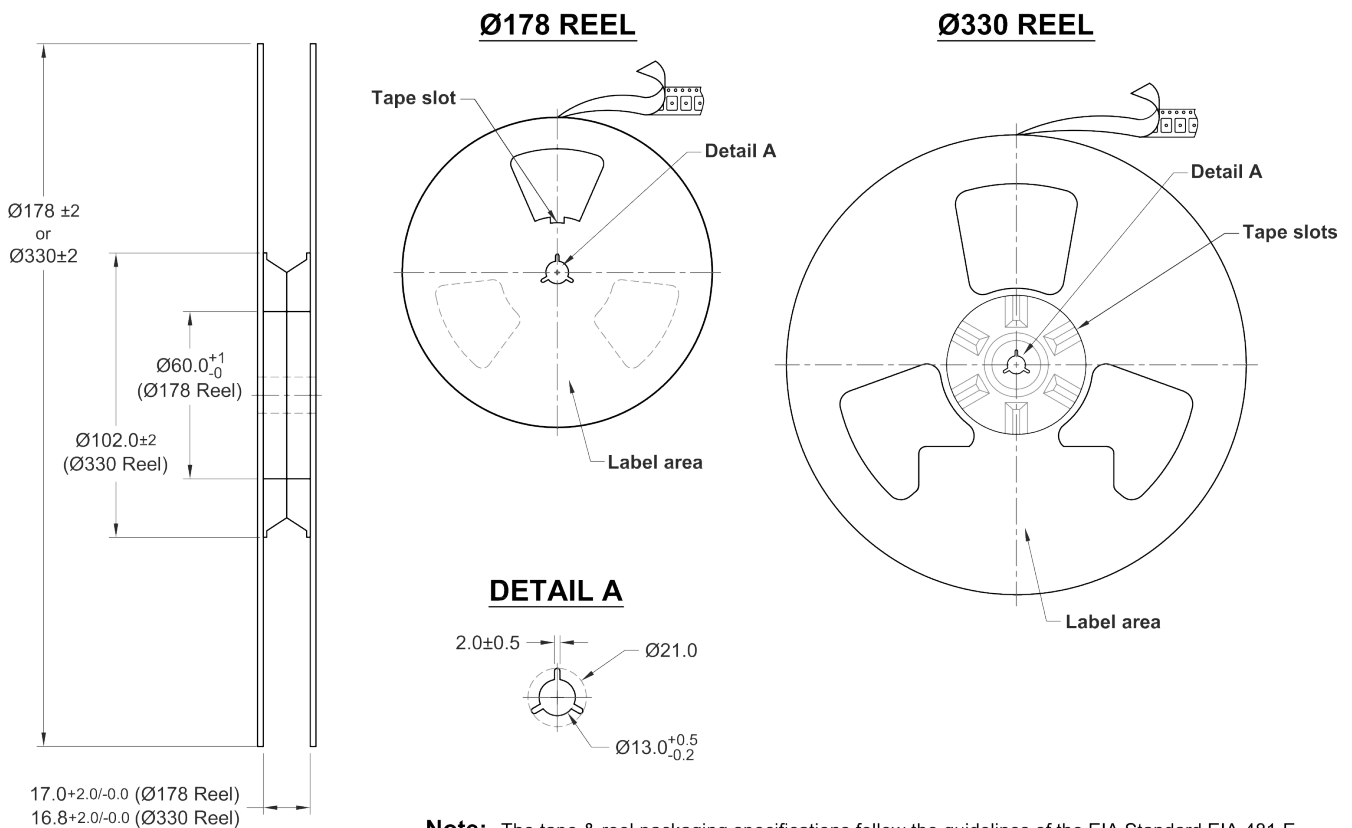
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15.0 Tape and Reel (Ø178mm/ Ø330mm):

TAPE DETAILS



REEL DETAILS



Note: The tape & reel packaging specifications follow the guidelines of the EIA Standard EIA-481-E.

TITLE: PLUTO/+/+2 7050 SERIES TAPE & REEL

RELATED DRAWINGS:

FILENAME: CAT778

REVISION: D

DATE: 22-Sep-2020

SCALE:

Millimetres

TOLERANCES:

XX = ± 1
$$\begin{aligned} X.X &= \pm 0.2 \\ Y.YY &= \pm 0.10 \end{aligned}$$

X.XX = ± 0.10
X.XXX =

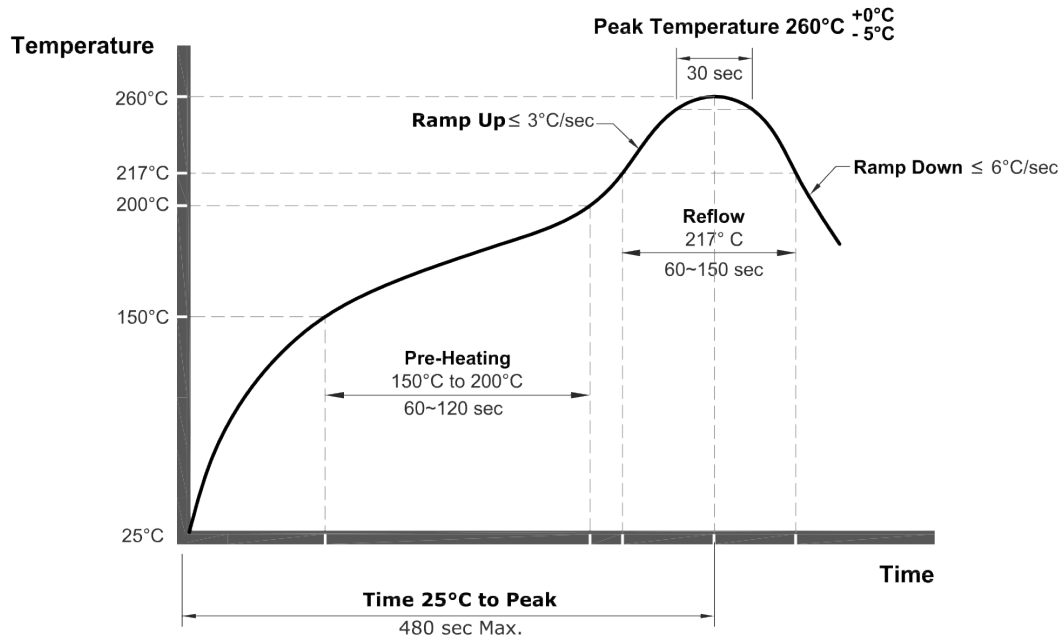
$$\chi^0 =$$

Hole =

rakon

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16.0 Reflow:



Note:

- The Pb-free Reflow follows the guidelines of IPC/JEDC J-STD-020E.
- The product has been tested to withstand the Reflow Profile shown. The Reflow Profile used to solder Rakon products is determined by the solder paste Manufacturer's specification. It is recommended that the Reflow Profile used does not exceed the one shown above.

TITLE: Pb-Free Crystal & Oscillator Reflow (Classification Temperature $T_c = 260^{\circ}\text{C}$)

FILENAME: CAT541

RELATED DRAWINGS:

REVISION: C

DATE: 16-May-2019

SCALE: NTS

Millimetres



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17.0 Specification History

Version	User	Changes	Approver	Date
A	Rock Steele	Specification created to replicate E8655LF	BC/ JM	2022-09-15
B	JO	Phase noise + pin 1 label corrected + minor typographical changes	LL/JMa	2022-10-04